

onto flash. These are read-access and write-access times.

Whether you are using apps like WhatsApp, or imaging or recording videos, you are running all these on the flash memory of your smartphone. Because these activities are run simultaneously, it is very critical to have excellent read and write times on your device. In other applications, like the Internet of Things (IoT) gateway or a set-top box, it might not be so critical.

Playing just a movie might not be very demanding on the memory, but multiple usages, like imaging, music and other running applications, create a critical demand for the need for speed. Companies like SanDisk differentiate their product lines according to performance and endurance. Their iNAND products range from standard to ultra, and finally extreme for highest performing chips.

Understanding what capacity your project really needs

Industrial devices do not usually deal with memory-heavy multimedia files or videos, and hence can make do with cheaper lower-capacity TLC memory. Alternatively, in those cases where the industrial device requires higher performance, it can move on to get an SLC NAND memory instead of TLC.

On the other hand, customer requirements for smartphones and consumer electronics are growing to 128GB for smartphones launched this year. In these cases, TLC is the better choice over SLC because the designer can access higher capacities of memory with the same budget constraint. Moreover, the end-user will value the size of the memory on the device quite heavily while making a purchase decision.

Memory capacity requirements are also based on the kind of operating system (OS) that you intend to run on the device. Microkernels are designed to work using minimal system resources and can be used with limited memory capacity. On the other hand, a

What does 3D NAND mean for the user?

3D NAND adds a third dimension to the traditional X and Y. This allows the technology to continue capacity and cost scaling, and outperform previous generations of NAND technology.

The biggest benefit made available to users is the increase in capacity. This technology has enabled the creation of SSDs that can store up to one terabyte of memory on a single drive.

Due to the removal of cell-to-cell interference from this picture, the programming algorithm that is implemented for storage can be a far simpler version of what is currently being used.

Finally, the use of an insulator, rather than a conductor, as the material used has had an effect on endurance. The insulator is far less prone to wear and tear than a conductor, and it is claimed that this can result in an endurance that is ten times higher.

ARM memory management

Instruction sets of ARMv8-A architecture come with new instructions for managing memory protection, cyclic redundancy check (CRC) instructions and enhanced floating-point instructions, according to the latest IEEE standard. ARMv8-R architecture specification will have high-end memory-protection capabilities with real-time and industrial-safety characteristics.

Types of Memory and Their Applications

Memory types	Brief description	Typical end applications
SSD	Solid-state drive, primary storage media based on semiconductor, unlike magnetic drives (hard drives); faster memory access, less power usage is the main USP	Personal computing devices (but not in desktops due to price being high); widely seen in tablets, laptops
NAND	Flash-type but not for random access, used for file access only (data storage only)	MP3 players, digital cameras, digital photo stands, PC cards, compact flash, secure digital media
NOR	Flash-type for random access (code storage only)	Home appliances, low-end consumer items (like mobiles—old generation)
USB	Universal serial bus, usually in flash drive format	Usually available as external memory agent to be used for all consumer items
SAN	Storage attached network, used to provide access to block-level access	E-mail servers, very frequent and high usage file servers (enterprise level)
NAS	Network attached storage was invented before SAN	Used in clusters for file system management
HDD	Traditional hard disk drive	Used in desktops, traditional computing, thin clients
HMC	Hybrid memory cubes, next-gen memories	Can be used in space-critical applications, routers

“Today most embedded NAND is now made to eMMC specs. This shift made it possible for vendors to make better decisions on selecting memory without worrying about rewriting the entire software”

— Vivek Tyagi, director, business development, OEM and enterprise, SanDisk India

full-featured OS requires more memory and plentiful system resources.

For every embedded design, software considerations are very important. “However, the term embedded

design is itself very broad, and so careful consideration has to be taken by the architect or system designer. The designer has to think on the amount of writing to be done. So, in